Year 12 Maths - Applications of differentiation

Similar questions - Tangents and normals

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The curve C has equation $y = x^2 - 4x + 7$. P is the point on C with x-coordinate t.

(a) Show that the tangent to C at P has equation $y = (2t-4)x + 7 - t^2$. **[4]**

(b) The tangent at P passes through the point $(0,3)$. Find the two possible values of t and the equation of each tangent. **[4]**

(c) Find the x-coordinate of the point where the tangent with negative gradient crosses the x-axis. **[1]**

2 The line $x + 3y = 20$ is the normal to the curve $y = ax^2 + \dfrac{b}{x}$ at the point $(2,6)$, where a and b are constants.

(a) Show that $8a + b = 12$. **[2]**

(b) Find the gradient of the curve at $(2,6)$ and hence show that $16a - b = 12$. **[3]**

(c) Find the values of a and b. **[2]**

Question 1

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2x-4$, so at $x=t$ the gradient is $2t-4$. $y-(t^2-4t+7)=(2t-4)(x-t)$, so $y=(2t-4)x-2t^2+4t+t^2-4t+7=(2t-4)x+7-t^2$

(b) Put $x=0, y=3$: $3=7-t^2$, so $t=\pm 2$. $t=2$: $y=3$. $t=-2$: $y=-8x+3$

(c) $-8x+3=0$ gives $x=\frac{3}{8}$

Question 2

(a) The curve passes through $(2,6)$: $4a+\frac{b}{2}=6$, so $8a+b=12$

(b) Normal $y=-\frac{1}{3}x+\frac{20}{3}$ has gradient $-\frac{1}{3}$, so the curve's gradient is 3. $\dfrac{\mathrm{d}y}{\mathrm{d}x}=2ax-bx^{-2}$ gives $4a-\frac{b}{4}=3$, so $16a-b=12$

(c) Adding: $24a=24$, so $a=1$ and $b=4$